UbiMyTherapist: A Digital Twin MultiModal LLM-based System with Emotion Detection

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***Abstract*—The integration of artificial intelligence (AI) in mental health therapy has led to the emergence of AI-powered therapeutic assistants. However, existing systems primarily function as reactive conversational agents, lacking real-time emotional context awareness. This paper presents UbiMyTherapist, a digital-twin, ubiquitous, multimodal framework that integrates a Retrieval-Augmented Large Language Model (RAG-LLM) with near-real-time emotion detection to provide both reactive and proactive support on consumer electronics such as smartwatches and smartphones. The framework builds a continuously updated digital-twin of the user's emotional state and medical history, leveraging affect recognition models that can process bio-signals, speech intonation, or text sentiment. This integration enables a more responsive AI therapy assistant. In this paper, we describe the system architecture, its components, and our prototype. The reactive mode was evaluated in a user study with 24 participants, where UbiMyTherapist outperformed baseline LLM setups on therapist-likeness and conversation quality.**

***Index Terms*—Digital-Twin, Ubiquitous biofeedback, RAG, Emotion detection, Generative AI, E-Health, Mental wellbeing**

I. Introduction

The integration of AI into mental health care can potentially transform the psychotherapy field due to its ability to enhance accessibility and continuous patient monitoring [1] [2]. One of AI's key strengths is its ability to analyze patient data, such as behavioral patterns, responses to treatment, and medical history [3], increasingly collected through consumer electronics such as wearables, and remote monitoring tools [4]. This is achieved through tools that utilize natural language processing (NLP), speech affect analysis, and other affective computing techniques. Leveraging these tools enables therapy sessions to be more personalized and tailored to each patient's unique needs and circumstances [5]. Some of these applications offer AI-powered virtual therapists [6] [7], which creates a safe and private space for people to find support and help reduce barriers that hinder individuals from seeking professional help [8] [9]. However, most such systems remain limited to user-initiated therapeutic interactions within the framework of conversational health agents [9] [10] [11]. These AI systems primarily respond to verbal or written communication but do not proactively engage in delivering mental health support, making them reactive rather than proactive. Moreover, the existing systems neglect real-time and continuous data analysis, which poses a significant drawback, particularly when immediate intervention is crucial. While chat-based therapy applications remain reactive, some AI-driven systems take a more proactive approach by incorporating physiological measurements for stress monitoring. For example, commercial applications such as EliteHRV [12] and CuesHub [13] focus on enhancing personal well-being by monitoring and analyzing data from wearables, proactively sending stress alerts, and suggesting relaxation techniques when high stress is detected. In a recent study exploring the integration of wearable devices with LLM for stress management, Neupane et al. [14] integrated CuesHub with LLM chatbots, offering proactive prompts, but their focus remained on stress reduction rather than structured therapy. Although these applications offer some aspects of proactive interventions, their primary focus is on stress reduction rather than delivering structured therapy sessions.

The Ubiquitous Biofeedback Multimedia Systems concept, first introduced by Al Osman et al. in 2014 [15], provides a foundation for proactive intervention through physiological measurements. A key part of these systems guides users through personalized relaxation practices by assessing their physiological measurements and adjusting interventions based on past effectiveness. More recently, the concept of digital-twin has emerged as a way to utilize emotional intelligence (EI) to enhance the quality of life and well-being [16]. In the context of consumer health, digital-twins are particularly effective when coupled with wearable devices, which generate continuous personal data. They provide the enabling technology that allows for continuous monitoring, deeper understanding, and timely feedback to improve well-being, and hold significant potential for application in the mental health field [17].

Building upon these two concepts, our proposed Digital-Twin Ubiquitous AI Personal Therapist Assistant (UbiMyTherapist) extends beyond traditional conversational chatbots by integrating agents and RAG-LLMs with emotion detection models that can operate on bio-signals, speech intonation, or text sentiment. Unlike other AI-powered systems, our framework does not rely solely on user interactions. It continuously monitors the user's emotional state through consumer electronics, such as smartwatches,

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smartphones, and earbuds, and delivers timely and appropriate interventions, even without user prompts. This digital-twin-driven, multimodal, and proactive approach is designed to ensure individuals receive dynamic and personalized assistance based on both their near-real-time emotional conditions and their historical context.

Unlike existing systems that rely on conversational LLMs, UbiMyTherapist uniquely integrates a digital-twin of the user's emotional history with near-real-time signals to deliver both reactive and proactive interventions. This paper presents the proposed framework's architecture, discussing its key components.

II. System Design and Overview

As illustrated in Fig. 1, the framework is composed of three layers: (1) Data Sources, (2) AI Inferencing Engine, and (3) Multimodal Display. The Data Sources layer collects information from both user-initiated interactions and continuous measurements captured by consumer electronics, while also managing knowledge stored in dedicated databases. The AI Inferencing Engine layer processes these inputs through different emotion recognition models to estimate near-real-time and historical emotional states. It also features a Generative AI module that integrates information retrieved by the RAG from databases, detected emotional states, and the user's prompt, if available, to produce personalized therapeutic responses using an LLM. Finally, the Multimodal Display layer determines the system interaction with the user, supporting text, voice, and haptic interfaces to deliver responses in two complementary modes: Reactive Mode, which acts as a conversational health agent, and Proactive Mode, which is based on ubiquitous biofeedback, enabling interaction without the user explicitly seeking help.

A. Data Sources

The system gathers data from three primary sources: (1) user-initiated inputs, (2) continuous data from consumer electronics, and (3) stored databases. The user-initiated inputs include video, speech, and text, which allow direct interaction with the system in a conversational setting. On the other hand, continuous data is collected passively from consumer electronics, such as smartwatches, smartphones, and other wearable devices. These devices can capture PPG, Electrocardiography (ECG), Electroencephalography (EEG), and Skin Conductance Response (SCR) through built-in sensors. Their portability and unobtrusiveness allow seamless monitoring of the user's physiological state without disrupting daily activities. The third source, the databases, plays a crucial role in supporting the digital-twin of the user by contextualizing and personalizing therapeutic responses:

1) *User's Medical History Database:* stores personal and medical information like age, gender, mental health diagnoses, and regular medications.
2) *Public Psychology Database:* Contains psychology books and literature. Instead of relying solely on the LLM's pre-trained parameters, our RAG module retrieves information from this source. This method improves therapeutic responses by ensuring they are fact-checked and contextually accurate.
3) *User's Emotional States Database:* Maintains a bi-directional interaction with the AI Inferencing Engine. Outputs from Emotion Recognition Models are continuously stored here, creating a historical record of the user's affective state. The RAG retriever later queries this database to improve therapeutic responses, supporting the digital twin concept with emotional data history.

B. AI Inferencing Engine

1) *Emotion Recognition Models:* This framework includes three models: Biosignals-to-Emotion, Speech-to-Emotion, and Text-to-Emotion, which can be used separately or combined through a decision strategy that weighs predictions by reliability and context for accurate emotional state estimation. The biosignals model processes physiological signals such as PPG, ECG, EEG, or SCR to predict emotions using machine learning. The second model analyzes speech input, focusing on vocal traits like tone, pitch, rhythm, and intensity. The third, Text-to-Emotion model, uses NLP to identify emotions in the user's text. Their outputs support both historical and real-time analysis and are stored in the User's Emotional States Database, which tracks emotional history over adjustable timescales. The RAG retriever integrates this context with knowledge from other databases, enabling responses to adapt in both content and style (e.g., tone and empathy), mirroring human therapists.
2) *Generative AI module:* This component features the RAG architecture, combining a retriever and a MultiModal Large Language Model (MM-LLM). The retriever queries databases for relevant information based on semantic similarity to the user's prompt, prioritizing meaning over exact wording. After retrieving content and emotional state, the MM-LLM generates a personalized response aligned with the user's mental and emotional condition. The response is delivered either reactively or proactively, depending on the interaction mode.

C. Multimodal Display

The Multimodal Display layer in Fig. 1 controls how UbiMyTherapist provides feedback via text, voice, and haptic interfaces, adapting to the user's emotional state to enhance engagement. For example, text responses offer detailed advice and coping tips within the App, voice responses deliver conversational guidance, such as breathing exercises, through earbuds, and haptic feedback, which subtly alerts users to stress, encouraging them to take deep breaths. By offering flexible interaction methods, the framework ensures responses remain contextually appropriate across both reactive (conversational) and proactive (biofeedback-triggered) modes

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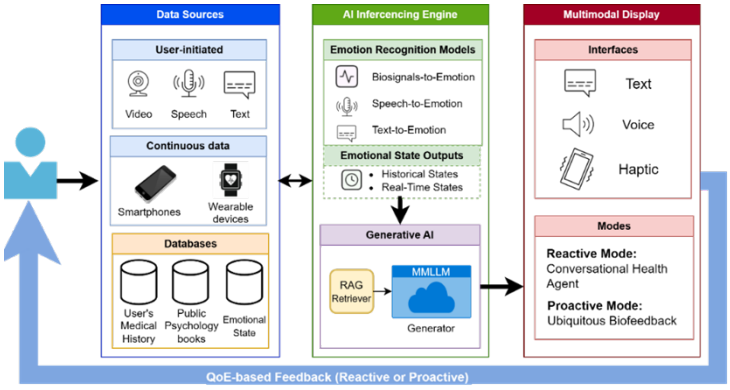


Fig. 1. UbiMyTherapist Framework Architecture, composed of three layers: Data Sources, AI Inferencing Engine, and Multimodal Display.

D. Feedback Modes

- **Reactive Mode (Conversational Health Agent):** In this mode, the user initiates interaction via text or voice, and the framework responds in a structured, personalized, and therapist-like manner.
- **Proactive Mode (Ubiquitous Biofeedback):** In this mode, the system autonomously detects emotional distress from real-time physiological signals collected by consumer electronics. When a negative state is identified, it proactively suggests interventions such as relaxation techniques, motivational prompts, or behavioral adjustments tailored to context. For example, while driving, the system may deliver short voice prompts for breathing exercises, whereas at home it could propose guided meditation or extended therapy sessions. By grounding decisions in the user's emotional state, the proactive mode implements the digital-twin concept, ensuring support is timely, non-intrusive, and contextually appropriate, even when the user does not explicitly seek help.

III. Prototype and Evaluation

To demonstrate the effectiveness of our proposed framework, we developed a prototype to evaluate the reactive mode.

A. Participants

The study involved 24 participants (10 male and 14 female) between the ages of 19 and 66 (Mean Average = 28.5). To capture a range of perspectives, participants differed in their prior use of LLMs: some reported daily use, others occasional use (at least once per week), and some had no prior experience. Educational backgrounds were also mixed, with 50% holding or pursuing graduate studies and 50% enrolled at the undergraduate level (50% computer-related fields, 29.2% psychology field, and 20.8% other fields). To assess familiarity with therapy, participants indicated whether they had previously attended therapy sessions, were aware of therapy practices but had not participated, or had no prior knowledge of therapy. This diversity in background and experience was intended to broaden the representativeness of the case study sample. The study was conducted after ethics approval was granted from the Research Ethics Board at the university of Ottawa.

B. Study protocol

The purpose of this study was to evaluate whether the reactive mode of UbiMyTherapist enhances the quality and therapist-likeness of conversational responses compared to baseline LLM configurations. Specifically, we asked: (1) *Which system produces responses most similar to those of a human therapist?* and (2) *How do participants perceive the realism, empathy, and conversational quality of different systems in a therapy-oriented context?*. Prior to the user study, we benchmarked several state-of-the-art LLMs (OpenAI GPT, Anthropic Claude, Google Gemini, and DeepSeek) to determine which one offered the most natural and therapist-like baseline. GPT-4o was selected as the base model for the case study. Participants then interacted with four virtual therapist configurations:

- System 1 - *GPT-4 baseline:* the raw model without modifications.
- System 2 - *GPT-4o + prompt Engineering:* baseline in addition to using a carefully designed Cognitive Behavioral Therapy (CBT)-style prompt to instruct the LLM to act as a therapist.
- System 3 - *RAG-based GPT-4o:* combining the same CBT prompt with retrieval from the Psychology Database described in Section II-A.
- System 4 - *UbiMyTherapist:* our full framework, integrating RAG with all three Data Sources (User's Medical History, Psychology Database, and User's Emotional State), in addition to the same CBT-style prompt.

To minimize bias, the order of system presentation was randomized for each participant. Each participant conducted short interactive sessions (5 to 10 minutes per system) and then completed a post-interaction questionnaire. The questionnaire contained four 5-point Likert items measuring Therapy-style Conversation, Empathy & Supportiveness, Personalization, and Session Closure, as well as one item measuring Future Use. Participants completed the study independently via a web interface, ensuring that system responses were evaluated without researcher intervention.

IV. Results and Discussion

As shown in Table I, UbiMyTherapist (system 4) achieved the highest mean rating on all five questions of the user case study: Therapy-style Conversation (4.50 ± 0.51), Empathy & Supportiveness (4.58 ± 0.58), Personalization (4.58 ± 0.72), Session Closure & Follow-up (4.33 ± 0.76), and if users would use the system for therapy sessions in the future (4.30 ± 0.69). UbiMyTherapist also showed a generally smaller standard deviation, indicating a more consistent user experience across participants. We attribute this advantage to the fact that UbiMyTherapist consistently integrates both the user's query and contextual information from emotional state and medical history, enabling more coherent responses.

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This contextual grounding improved not only empathy and personalization but also secondary aspects such as session closure, which are not directly encoded in prompts. Even without the user's medical and emotional state history, the RAG-based system (system 3) outperformed system 2 (mean average 3.83 and 3.25, respectively), while GPT-4o baseline (system 1) scored the lowest (1.53 average). Adding prompt engineering to the GPT baseline substantially improved ratings across all criteria. Comparing System 1 and System 2, the mean scores increased on all five questions: Therapy-style Conversation (1.25 to 3.38), Empathy & Supportiveness (1.63 to 3.38), Personalization (1.75 to 3.38), Session Closure & Follow-up (1.83 to 3.30), and Future Use (1.20 to 2.83). Even though prompt engineering enhanced GPT-4o's performance, our digital-twin solution, which combines the power of RAG with continuous updates of the user's emotional state and medical history, significantly outperformed it. In our immediate future work, we will focus on evaluating the context, not just the user's perspective, by recruiting registered psychotherapists to provide expert evaluations of contextual accuracy and therapeutic soundness across systems. To assess whether differences among the four systems were statistically significant, we ran a non-parametric Friedman test for each question (this test is appropriate for ordinal Likert data in a within-subjects design). The results were significant for all five questions with a p-value less than 0.001 ($X^2(3, 24)$: 51.7, 51.1, 48.26, 43.6, and 51.7, respectively). These findings confirm that the reactive mode of UbiMyTherapist significantly outperformed the other systems.

TABLE I
USER EVALUATION OF THE FOUR VIRTUAL THERAPIST SYSTEMS

Mean ± SD	***System 1***	***System 2***	***System 3***	***System 4***
Therapy-style Conversation	1.25 ± 0.90	3.38 ± 1.01	3.92 ± 0.83	**4.50 ± 0.51**
Empathy and Supportiveness	1.63 ± 1.10	3.38 ± 1.06	4.04 ± 0.75	**4.58 ± 0.58**
Personalization	1.75 ± 0.99	3.38 ± 1.01	4.00 ± 0.72	**4.58 ± 0.72**
Session Closure and Follow-up	1.83 ± 1.00	3.30 ± 1.16	3.75 ± 1.03	**4.33 ± 0.76**
Using the system for therapy sessions	1.20 ± 0.83	2.83 ± 0.96	3.46 ± 1.02	**4.30 ± 0.69**

V. CONCLUSION AND FUTURE WORK

This paper introduces UbiMyTherapist, a digital-twin–based AI therapist assistant that integrates RAG-LLMs with multimodal emotion detection for more responsive and personalized support. The framework consists of three layers: Data Sources, AI Inferencing Engine, and Multimodal Display, which together combine the user's medical history, near-real-time biosignals, and psychological knowledge to generate context-aware responses. By leveraging consumer electronics, such as smartwatches and smartphones, UbiMyTherapist extends AI-driven therapy beyond traditional chat agents, offering both reactive and proactive feedback. A 24-participant case study demonstrated the effectiveness of the reactive mode, showing improved therapist-likeness and conversation quality when user context was incorporated. The key novelty lies in the digital-twin foundation, which continuously maintains a representation of the user's emotional history, enabling personalized and contextually relevant interventions. While our current evaluation was limited to the reactive mode, the results provide promising evidence of feasibility. Future work will focus on three directions: (1) completing the proactive integration using our validated PPG-based model [18] to trigger interventions in near real-time, (2) collaborating with licensed psychotherapists to evaluate therapeutic soundness and ensure clinical alignment, and (3) expanding to additional modalities such as speech, facial expressions, and contextual sensing to enrich the digital twin.

REFERENCES

[1] A. Mittal, L. Dumka, and L. Mohan, "A comprehensive review on the use of artificial intelligence in mental health care," in *14th ICCCNT*. IEEE, 7 2023, pp. 1–5.

[2] L. Yu, L. Wang, J. Cai, Z. Yang, L. Wen, A. K. Bashir, and W. Wang, "Consumer electronics and genai providing user experiences in mental health," *IEEE Consumer Electronics Magazine*, pp. 1–9, 2024.

[3] S. Iqbal, X. Zhong, M. A. Khan, M. Shabaz, Z. Wu, D. A. AlHammadi, W. Liu, S. A. Algamdi, and Y. Li, "Transforming healthcare diagnostics with tensorized attention and continual learning on multi-modal data," *IEEE Transactions on Consumer Electronics*, 2025.

[4] X. Wang, Z. Liu, L. Zou, J. Wang, X. Zhang, and N. Liu, "Large-scale medical records analysis by ai-driven method in healthcare consumer electronics," *IEEE Transactions on Consumer Electronics*, vol. 71, pp. 1463–1472, 2 2025.

[5] S.-H. Lee, T.-Y. Chen, Y.-T. Hsien, and L.-R. Cao, "A music recommendation system for depression therapy based on eeg," in *International Conference on Consumer Electronics-Taiwan (ICCE)*. IEEE, 2020.

[6] "Ellie mental health - elliementalhealth.com."

[7] T. Dong, F. Liu, X. Wang, Y. Jiang, X. Zhang, and X. Sun, *EmoAda: A Multimodal Emotion Interaction and Psychological Adaptation System*, 2024.

[8] A. Husnain, A. Ahmad, A. Saeed, and S. M. U. Din, "Harnessing ai in depression therapy: Integrating technology with traditional approaches," *International Journal of Science and Research Archive*, 2024.

[9] K. K. Fitzpatrick, A. Darcy, and M. Vierhile, "Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (woebot): A randomized controlled trial," *JMIR Mental Health*, 2017.

[10] B. Inkster, S. Sarda, and V. Subramanian, "An empathy-driven, conversational artificial intelligence agent (wysa) for digital mental well-being: Real-world data evaluation mixed-methods study," *JMIR mHealth and uHealth*, vol. 6, p. e12106, 11 2018.

[11] A. Mehta, A. N. Niles, J. H. Vargas, T. Marafon, D. D. Couto, and J. J. Gross, "Acceptability and effectiveness of artificial intelligence therapy for anxiety and depression (youper): Longitudinal observational study," *Journal of Medical Internet Research*, vol. 23, p. e26771, 6 2021.

[12] "Elitehrv - elitehrv.com," *elitehrv.com*.

[13] "Cueshub - cueshub.com," *https://www.cueshub.com/*.

[14] S. Neupane, P. Dongre, D. Gracanin, and S. Kumar, "Wearable meets llm for stress management: A duoethnographic study integrating wearable-triggered stressors and llm chatbots for personalized interventions," 2025.

[15] H. A. Osman, M. Eid, and A. E. Saddik, "U-biofeedback: a multimedia-based reference model for ubiquitous biofeedback systems," *Multimedia Tools and Applications*, 2014.

[16] A. Albraikan, B. Hafidh, and A. E. Saddik, "Iaware: A real-time emotional biofeedback system based on physiological signals," *IEEE Access*, 2018.

[17] N. Bagaria, F. Laamarti, H. F. Badawi, A. Albraikan, R. A. M. Velazquez, and A. E. Saddik, *Health 4.0: Digital twins for health andwell-being*. Springer International Publishing, 1 2019, pp. 143–152.

[18] K. Alghoul, H. A. Osman, and A. E. Saddik, "Enhancing generalization in ppg-based emotion recognition with a cnn-tcn-lstm model," in *I2MTC*. IEEE, 2025.

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